Data Science Mids Guide

Mid-term topics: Weeks 1–7 and Week 10 of the lecture plan. Easy notes with Urdu examples, live graphs, scenario practice, quizzes and a Python code judge.

WEEK 1 (TOPIC 1 OF 8)

Introduction to Data Science

Definition, history, applications, how it differs from AI, ML and analytics, the lifecycle and the roles in a data team.

1 What is Data Science?

KEY POINT

Data Science means getting **useful insights from raw data**. It mixes **statistics**, **programming** and **domain knowledge**.

- Statistics: understand patterns and uncertainty.
- Programming: collect, clean and analyse data at scale (Python, R, SQL).
- Domain knowledge: know what the numbers mean for the real problem.

Exam definition: Data science is an interdisciplinary field that combines mathematics, statistics, computer science, artificial intelligence and domain expertise to extract meaningful insights and knowledge from raw, structured or unstructured data.

اردو مثال

ڈیٹا سائنس کا مطلب خام ڈیٹا سے کارآمد معلومات نکالنا ہے۔ جیسے دکاندار پچھلے سالوں کی فروخت دیکھ کر اندازہ لگاتا ہے کہ عید پر کون سا سامان زیادہ بِکے گا۔ اس میں شماریات، پروگرامنگ اور اس شعبے کی سمجھ، تینوں شامل ہوتی ہیں۔

2 History and development

When	Milestone (commonly cited)
1962	John Tukey argues for a science of **data analysis** in “The Future of Data Analysis”.
1974	Peter Naur uses the term **“data science”** for the science of dealing with data.
1997–2001	Statisticians (C. F. Jeff Wu, William Cleveland) propose data science as an extension of statistics with computing.
2000s	Growth of the web, large-scale data tools (Hadoop, 2006) and cheap cloud storage make analysis of huge datasets possible.
2008–2012	The job title **“data scientist”** becomes popular; deep learning and machine learning advance quickly.
Today	Data science is used everywhere, with AI, automation and cloud platforms.

> ⚠ **NOTE**
> These are widely quoted milestones. Check your lecture slides for the dates and names your teacher stresses.

> 💡 **KEY POINT**
> Short story: **statistics** → **computing** → **large-scale data** → **machine learning and AI**.

3 Applications of Data Science

Field	Example
Healthcare	Predicting disease risk from test results
Banking and finance	Detecting fraudulent card transactions; credit scoring
E-commerce and media	Recommending products or videos
Education	Predicting student performance
Real estate	Predicting house prices
Transport	Route planning, estimating arrival time
Agriculture	Predicting crop yield from weather and soil data
Manufacturing	Predicting machine failures

4 Data Science vs AI vs ML vs Data Analytics

Term	Meaning	Example
Artificial Intelligence (AI)	Machines doing tasks that normally need human intelligence	A chatbot, a self-driving car
Machine Learning (ML)	A **subset of AI**: algorithms learn patterns from data instead of following fixed rules	Spam filter learned from labelled emails
Data Science (DS)	Interdisciplinary field that uses statistics, programming, domain knowledge (and often ML) to get insight from data	Predicting student performance end to end
Data Analytics (DA)	Examining (mostly past) data and presenting it clearly: reports, dashboards	Monthly sales dashboard

> 💡 **KEY POINT**
> **AI** ⊃ **ML**. Data science **uses** ML (and statistics) but is broader: it covers the whole path from raw data to insight.

5 The Data Science lifecycle

> 💡 **KEY POINT**
> A lifecycle is a step-by-step process: **Understand → Acquire → Explore → Clean → Analyse → Engineer → Model → Evaluate → Deploy → Monitor**.

#	Stage	What you do
1	Problem / business understanding	Define the problem, objective, target variable, constraints and success criteria. (“Pehle samjho kya solve karna hai.”)
2	Data acquisition (collection)	Collect data from databases, APIs, files, sensors, websites, surveys.
3	Data understanding / exploration	Inspect structure, variables, distributions, missing values, duplicates.
4	Data cleaning / preparation	Fix missing values, duplicates, wrong formats, outliers, inconsistent data.
5	EDA and analysis	Use statistics and charts to discover patterns and relationships.
6	Feature engineering / selection	Create useful features and select relevant variables.
7	Model building (ML modelling)	Train machine-learning or statistical models.
8	Model evaluation	Test the model on **unseen rows** using suitable measures.
9	Deployment	Put the model into practical use (make it live).
10	Monitoring and maintenance	Track performance, data drift, errors and when to retrain.

> ⚠️ **NOTE**
> Your handwritten notes list a shorter version (problem definition, data collection, data cleaning, EDA, ML modelling, evaluation, deployment and monitor). Frameworks may use different numbers or names of stages; the order of ideas is the same.

اردو مثال

فرض کریں آپ جاننا چاہتے ہیں کہ کون سے طلبہ فیل ہو سکتے ہیں۔ پہلے مسئلہ طے کریں، پھر حاضری اور نمبروں کا ڈیٹا جمع کریں، غلط یا خالی خانے ٹھیک کریں، چارٹ بنا کر پیٹرن دیکھیں، ماڈل بنائیں، نئے طلبہ پر اس کی جانچ کریں اور آخر میں اسے استعمال میں لا کر نگرانی کرتے رہیں۔

6 Roles and responsibilities in Data Science

> 💡 **KEY POINT**
> Data Scientist = **models and statistics**. Data Analyst = **reports and visualisation**. Data Engineer = **pipelines**. ML Engineer = **deploys models**.

Role	Main job	Tools (from your notes)
Data Scientist	Modelling and statistics; analyses data and builds predictive models	Python, R, SQL, TensorFlow
Data Analyst	Reporting and visualisation; dashboards	Excel, SQL, Tableau, Power BI
Data Engineer	Builds data pipelines and infrastructure	Spark, Hadoop, Airflow, Kafka
ML Engineer	Deploys models to production and maintains ML systems	PyTorch, TensorFlow, Docker
Data Architect	Designs data architecture and integration	Databases, cloud, architecture
Business / Domain Analyst	Connects business needs with data solutions	Domain knowledge, communication

The **value** of all these roles is **business insight** from data. Roles overlap in real teams, but each has a main responsibility.

اردو مثال

ہسپتال میں: ڈیٹا انجینئر مریضوں کے ریکارڈ کی پائپ لائن بناتا ہے، ڈیٹا اینالسٹ ماہانہ رپورٹ بناتا ہے، ڈیٹا سائنٹسٹ بیماری کی پیشگوئی کا ماڈل بناتا ہے اور ایم ایل انجینئر اسے اصل نظام میں چلاتا ہے۔

📝 Mid-exam style questions (Week 1)

Define Data Science and explain the lifecycle.

Differentiate Data Science, AI, ML and Data Analytics.

Explain the roles of Data Scientist, Analyst, Engineer and ML Engineer.

Why is monitoring needed after deployment?

WEEK 2 (TOPIC 2 OF 8)

Python for Data Science

Python syntax, data types, collections, conditions, loops, functions and NumPy. All code on this page was run to check its output.

1 Python introduction, syntax and Google Colab

> 💡 **KEY POINT**
> Python is a readable programming language. **Indentation** (spaces at the start of a line) shows which lines belong together. **#** starts a comment.

- **Google Colab** gives free Python notebooks in the browser, with nothing to install (Lab 1).
- The notes call Python “a full workshop”: you can build anything with libraries like pandas, seaborn and scikit-learn.

```
print("Hello, Data Science")
# this line is a comment
name = "Ali"
print("Hello", name)
```

Output: Hello, Data Science / Hello Ali.

2 Variables and data types

Type	Meaning	Example
int	whole number	`20`
float	number with a decimal point	`1.75`
str	text	`"Sara"`
bool	True or False	`True`

```
age = 20
height = 1.75
name = "Sara"
passed = True
print(type(age), type(height), type(name), type(passed))
```

type() tells you the data type of a variable.

3 Collections: list, tuple, set, dictionary

Collection	Written	Ordered?	Changeable?	Notes
list	`[78, 91, 55]`	yes	yes	duplicates allowed; index from 0
tuple	`(31.5, 74.3)`	yes	no	fixed once created
set	`{1, 2, 3}`	no	yes	only unique items
dictionary	`{"name": "Ali"}`	keeps insertion order	yes	key → value; keys are unique

```
marks = [78, 91, 55]
print(marks[0], marks[-1])
student = {"name": "Ali", "marks": 78}
print(student["name"])
print(len(set([1, 2, 2, 3])))
```

Index 0 is the first item and -1 is the last. A set removes the repeated 2.

4 Operators and conditions

- **Arithmetic:** + − * / and // (whole division), % (remainder), ** (power).
- **Comparison:** == != > < >= <=
- **Logical:** and, or, not

```
marks = 72
if marks >= 80:
    print("A")
elif marks >= 60:
    print("B")
else:
    print("C")
print(7 // 2, 7 % 2, 2 ** 3)
```

72 is not ≥ 80 but is ≥ 60, so it prints B. 7 // 2 = 3, 7 % 2 = 1, 2 ** 3 = 8.

5 Loops and functions

- **for** repeats over a sequence (a list, or `range(1, 5)` = 1, 2, 3, 4).
- **while** repeats while a condition stays True. Make sure it can end!
- **Functions** (`def`) package reusable code; `return` sends back a result.

```
total = 0
for v in [1, 2, 3]:
    total += v

def average(nums):
    return sum(nums) / len(nums)

print(total, average([2, 4, 6]))
count = 0
while count < 3:
    count += 1
print(count)
```

> 💡 **KEY POINT**
> Your homework task: a Python program using **an array, a loop, conditions and a calculator**. Try the exercises in the Python judge.

6 NumPy: arrays and operations

> 💡 **KEY POINT**
> NumPy is the library for **numerical computing**. Its array holds numbers of one type and does maths on **all elements at once**.

```
import numpy as np
a = np.array([10, 20, 30, 40])
print(a * 2)
print(a.mean(), np.median(a), a.sum(), a.min(), a.max())
print(a[1:3])
```

Key functions from your notes: np.array(), np.mean(), np.dot(). Install with `pip install numpy`. Slicing a[1:3] gives items at index 1 and 2.

	Python list	**NumPy array**
Types	can mix types	one type (fast, compact)
+	joins lists: [1, 2] + [3] → [1, 2, 3]	adds element-wise: np.array([1, 2]) + 3 → [4 5]
* 2	repeats the list	doubles every element
Maths	needs loops	vectorised: one expression, many numbers

```
import numpy as np
print([1, 2] + [3])
print(np.array([1, 2]) + 3)
```

7 Python libraries for Data Science

Library	Purpose	Key functions (from your notes)
NumPy	Numerical computing, arrays	`np.array(), np.mean(), np.dot()`
Pandas	Data manipulation, DataFrames	`pd.read_csv(), df.groupby(), df.merge()`
Matplotlib	Basic plotting	`plt.plot(), plt.bar(), plt.scatter(), plt.show()`
Seaborn	Statistical visualisation	`sns.heatmap(), sns.boxplot(), sns.pairplot()`
Scikit-learn	Machine learning	`train_test_split()`

> 💡 **KEY POINT**
> NumPy = numerical · Pandas = data · Matplotlib / Seaborn = visualisation · Scikit-learn = ML.

📝 Mid-exam style questions (Week 2)

List the Python data types and collections.

Why use NumPy arrays instead of lists for numerical work?

Write a function that returns the average of a list.

WEEK 3 (TOPIC 3 OF 8)

Data Acquisition

Where data comes from, in what form, and how to judge whether it is good enough.

1 What is data acquisition?

> 💡 **KEY POINT**
> Data acquisition = **obtaining data** from relevant sources for analysis. Data itself is **raw facts and figures**.

Forms of data (from your notes): **pictorial** (images), **textual** (words) and **digital** (stored electronically).

Why it matters: the quality of the analysis can never be better than the quality of the data you start with.

2 Primary and secondary data

	Primary data	Secondary data
Meaning	Collected **directly** for the current purpose	Already collected by someone else
Examples (notes)	Interview, survey, questionnaire (also experiments, sensors)	Repository, database data, cloud (also government and research datasets)
Good	Fits your question exactly	Quick and cheap
Watch out	Costly and slow	May not fit your question; check quality and permission

اردو مثال

آپ خود طلبہ سے سوالنامہ بھروائیں تو یہ بنیادی (primary) ڈیٹا ہے۔ اگر آپ حکومت کی شائع کردہ مردم شماری کا ڈیٹا ڈاؤن لوڈ کریں تو یہ ثانوی (secondary) ہے۔

3 Structured, semi-structured and unstructured data

	Structured	Semi-structured	Unstructured
Shape	Fixed rows and columns	Flexible, uses keys / tags	No predefined table
Examples	SQL tables, student or bank database	JSON, XML	Images, text, audio, video
Typical tools	SQL, spreadsheets	JSON/XML parsers, APIs	Image / text / audio processing

Example of unstructured data use: digital image processing, like the filters in Snapchat.

اردو مثال

اسکول کا داخلہ رجسٹر (خانوں والا) ساختہ ڈیٹا ہے، کسی ایپ کی JSON فائل نیم ساختہ ہے اور موبائل میں رکھی تصویریں اور وائس نوٹس غیر ساختہ ڈیٹا ہیں۔

4 Data formats and sources: CSV, Excel, JSON, databases, APIs

Source	What it is	Reading it in Python
CSV	Plain-text table, values separated by commas (sales records)	`pd.read_csv("file.csv")`
Excel	Workbook with one or more sheets	`pd.read_excel("file.xlsx")`
JSON	Key-value data, common in web responses	`json.loads(text)` or `pd.read_json()`
Database	Organised tables queried with SQL (customer records)	SQL query, e.g. `SELECT * FROM students`
API	A web service that replies with data (often JSON) when your program asks	send a request, then read the JSON reply

```
import json
text = '{"name": "Ali", "marks": [70, 80]}'
data = json.loads(text)
print(data["marks"][1])

import pandas as pd
from io import StringIO
csv = "name,marks\nAli,78\nSara,91"
df = pd.read_csv(StringIO(csv))
print(df.shape)
```

json.loads turns JSON text into a dictionary. read_csv turns CSV text into a DataFrame with 2 rows and 2 columns (shape (2, 2)).

5 Concerns when acquiring data

- **Permission:** do you have the right to use it (terms of use, consent)?
- **Privacy:** protect personal details.
- **Quality:** accurate, complete, consistent?
- **Relevance and reliability:** does it answer your question, and can you trust the source?
- **Cost and format:** can you afford and read it?
- **Collection bias:** does the sample represent everyone?

6 Data quality dimensions

Dimension	Question	Example problem
Accuracy	Is it correct?	Age = 250
Completeness	Are values present?	Missing marks
Consistency	Same format everywhere?	“Lahore” vs “lahore”
Validity	Does it follow the rules?	Negative quantity
Uniqueness	No unwanted duplicates?	Same student twice
Timeliness	Is it up to date?	Last year's prices

📝 Mid-exam style questions (Week 3)

What is data acquisition? Explain primary and secondary data.

Differentiate structured, semi-structured and unstructured data.

Name the data quality dimensions.

WEEK 4 (TOPIC 4 OF 8)

Data Cleaning

Missing values, duplicates, inconsistent data, outliers, encoding and scaling. Raw data is almost never ready to use.

1 What is data cleaning? The workflow

> 💡 **KEY POINT**
> Data cleaning = detecting and correcting **inaccurate, incomplete, inconsistent, duplicate or irrelevant** data. Notes: fix weak or missing data; incorrect and duplicate values.

1. **Inspect** the data (shape, info, describe, null check).
2. Remove **duplicates**.
3. Handle **missing values**.
4. Fix **types** and **inconsistent** values (spelling, case, units).
5. Investigate **outliers**.
6. **Encode** categories and **scale** numbers if the model needs it.
7. **Check again**.

> 💡 **KEY POINT**
> Exam focus: cleaning is **not** simply deleting bad rows. First understand *why* a value is missing, wrong or unusual.

اردو مثال

رجسٹر میں ایک طالب علم کا نام دو بار لکھا ہے (ڈپلیکیٹ)، کسی کے نمبر خالی ہیں، 'Lahore' اور 'lahore' الگ لکھے ہیں اور کسی کی عمر 250 لکھی ہے۔ صفائی میں ان سب کو دیکھ کر ٹھیک کیا جاتا ہے۔

2 Missing values

Method	Use when
Delete rows	Very few rows are missing and the loss is small
Delete the column	Most of its values are missing
Fill with the mean	Numeric data without strong outliers
Fill with the median	Numeric data with outliers or skew
Fill with the mode	Categorical data (most frequent value)

Filling gaps with an estimate is called **imputation**.

```
import pandas as pd
s = pd.Series([70, None, 90, 80])
print(s.isnull().sum())
print(s.fillna(s.mean()).tolist())
```

One value is missing; the mean of 70, 90 and 80 is 80, so the gap becomes 80.0.

3 Duplicates and inconsistent data

- **Duplicates:** the same record twice. `df.drop_duplicates()` keeps the first copy.
- **Inconsistent values:** "Lahore", "lahore " and "LAHORE" are the same city written differently. Standardise spaces, case, units and spelling.

```
import pandas as pd
df = pd.DataFrame({"city": ["Lahore", "lahore ", "Karachi", "Lahore"]})
df["city"] = df["city"].str.strip().str.title()
print(df["city"].tolist())
print(len(df.drop_duplicates()))
```

After cleaning the text, only two distinct rows remain (Lahore, Karachi). Clean the text **before** looking for duplicates.

4 Outliers

> 💡 **KEY POINT**
> An **outlier** is a value far from the others. **Detect** it, then **investigate**: is it an error (fix it) or a real rare value (keep it)?

IQR rule: outlier if the value is below **Q1 − 1.5×IQR** or above **Q3 + 1.5×IQR**, where IQR = Q3 − Q1. Another idea: a very large |z-score| (for example above 3).

```
import pandas as pd
s = pd.Series([30, 32, 35, 36, 38, 40, 45, 160])
q1, q3 = s.quantile(0.25), s.quantile(0.75)
iqr = q3 - q1
print(q1, q3, s[(s < q1 - 1.5*iqr) | (s > q3 + 1.5*iqr)].tolist())
```

Q1 = 34.25, Q3 = 41.25, IQR = 7, upper fence = 51.75, so 160 is an outlier.

5 Variable types and categorical encoding

Type	Meaning	Example
Numerical: discrete	Counted whole numbers	Number of students
Numerical: continuous	Measured on a scale	Height, temperature
Categorical: nominal	Labels with **no** order	City, product category
Categorical: ordinal	Labels with a **meaningful order**	Low / medium / high
Boolean	True/False or 1/0	Owns a laptop

> 💡 **KEY POINT**
> Models need numbers. **Label encoding** gives each category a number (use for **ordinal** data). **One-hot encoding** makes a 0/1 column per category (use for **nominal** data, so no false order appears).

```
import pandas as pd
df = pd.DataFrame({"city": ["Lahore", "Karachi", "Lahore"]})
print(pd.get_dummies(df["city"], dtype=int).columns.tolist())
```

6 Feature scaling: normalisation and standardisation

> 💡 **KEY POINT**
> **Min-max normalisation:** (x − min) / (max − min) → values from 0 to 1.
> **Standardisation (z-score):** (x − mean) / SD → mean 0, SD 1.

Worked example with 20, 30, 40, 50, 100 (min 20, max 100, mean 48, SD ≈ 27.86): the age 40 scales to (40 − 20) / 80 = 0.25; the age 100 has z = (100 − 48) / 27.86 = 1.87.

```
x = [20, 30, 40, 50, 100]
lo, hi = min(x), max(x)
print([round((v - lo) / (hi - lo), 3) for v in x])
mean = sum(x) / len(x)
sd = (sum((v - mean) ** 2 for v in x) / len(x)) ** 0.5
print(round(mean, 1), round(sd, 2), round((100 - mean) / sd, 2))
```

> ⚠️ **NOTE**
> **Data leakage:** learn the min, max, mean and SD (and imputation values) from the **training data only**, then apply them to the test data.

اردو مثال

عمر 20 سے 100 تک اور تنخواہ لاکھوں میں ہو تو بڑے عدد والا فیچر ماڈل پر غالب آ جاتا ہے۔ اسکیلنگ سے سب کو ایک جیسے پیمانے پر لایا جاتا ہے۔

📝 Mid-exam style questions (Week 4)

What is data cleaning? List common problems and fixes.

Label encoding vs one-hot encoding?

Min-max normalisation vs standardisation?

WEEK 5 (TOPIC 5 OF 8)

Pandas & Data Manipulation

Series and DataFrames, reading and inspecting data, filtering, sorting, groupby, aggregation, merge and join.

1 Pandas introduction: Series and DataFrame

💡 KEY POINT

Pandas is the main Python library for data manipulation. A **Series** is one labelled column; a **DataFrame** is a table of rows and columns.

```
import pandas as pd
s = pd.Series([78, 91, 55], index=["Ali", "Sara", "Usman"])
print(s["Sara"])
df = pd.DataFrame({"name": ["Ali", "Sara"], "marks": [78, 91]})
print(df.shape)
```

s["Sara"] looks up the value by label. df.shape is (rows, columns) = (2, 2).

2 Reading and inspecting data

Task	Call
Read a CSV / Excel file	`pd.read_csv("f.csv"), pd.read_excel("f.xlsx")`
First rows	`df.head()`
Columns, types, non-null counts	`df.info()`
Summary statistics	`df.describe()`
Number of rows and columns	`df.shape`
Missing values per column	`df.isnull().sum()`
Remove repeated rows	`df.drop_duplicates()`

```
import pandas as pd
df = pd.DataFrame({"name": ["Ali", "Sara", None], "marks": [78, 91, 55]})
print(df.shape)
print(df.isnull().sum().to_dict())
```

3 rows and 2 columns; the name column has 1 missing value.

3 Filtering and sorting

> 💡 **KEY POINT**
> **Filtering** keeps rows where a condition is True: `df[df["marks"] > 60]`. **Sorting** orders rows: `df.sort_values("marks", ascending=False)`.

```
import pandas as pd
df = pd.DataFrame({"name": ["Ali", "Sara", "Usman"], "marks": [78, 91, 55]})
print(df[df["marks"] > 60]["name"].tolist())
print(df.sort_values("marks", ascending=False)["name"].tolist())
```

Only Ali (78) and Sara (91) pass the filter; sorted high to low gives Sara, Ali, Usman.

4 GroupBy and aggregation

> 💡 **KEY POINT**
> **groupby** = split the rows into groups, apply a function to each group, combine the results.

- Common aggregations: **sum, mean, count, min, max**.
- Group keys come out sorted by default.

```
import pandas as pd
df = pd.DataFrame({"city": ["Lahore", "Karachi", "Lahore"], "marks": [78, 91, 72]})
print(df.groupby("city")["marks"].mean().to_dict())
```

Lahore: (78 + 72) / 2 = 75.0. Karachi: 91.0.

5 Merge and join

> 💡 **KEY POINT**
> **merge** combines two tables through shared **key columns**. **join** combines tables aligning on the **index** by default. Both support several join types.

Join type	Keeps
inner (default for merge)	only rows whose key is in both tables
left	all rows of the left table, matched rows of the right
right	all rows of the right table, matched rows of the left
outer	all rows from both tables

```
import pandas as pd
a = pd.DataFrame({"city": ["Lahore", "Karachi"], "marks": [75, 91]})
b = pd.DataFrame({"city": ["Lahore", "Karachi"], "province": ["Punjab", "Sindh"]})
print(a.merge(b, on="city")["province"].tolist())
```

6 Data manipulation overview

> 💡 **KEY POINT**
> Manipulation = **transforming and organising** data so it can be analysed. Cleaning fixes quality problems; manipulation reshapes data for analysis.

- Filtering rows, selecting or removing columns
- Sorting values
- Grouping and aggregation (sum, mean, count)
- Merging / joining datasets
- Creating calculated (derived) columns, changing data types

Order in a project (notes): acquisition (get data) → cleaning (fix quality) → manipulation (Pandas) → visualisation (charts and graphs).

📝 Mid-exam style questions (Week 5)

Series vs DataFrame?

Explain groupby and aggregation with an example.

Difference between merge and join?

WEEK 6 (TOPIC 6 OF 8)

Exploratory Data Analysis (EDA)

Looking at the data before modelling: univariate, bivariate and multivariate analysis, descriptive statistics and correlation.

1 Meaning and purpose of EDA

> 💡 KEY POINT
> **EDA** (Exploratory Data Analysis) = explore the data to find **patterns, problems and relationships** before building a model.

From your notes, EDA always starts with: **shape** → **info** → **describe** → **null check**.

Step	Pandas call	Tells you
shape	`df.shape`	number of rows and columns
info	`df.info()`	column names, types, non-null counts
describe	`df.describe()`	count, mean, std, min, quartiles, max
null check	`df.isnull().sum()`	missing values per column

```
import pandas as pd
df = pd.read_csv("students.csv")   # needs your own file
print(df.shape)
df.info()
print(df.describe())
print(df.isnull().sum())
```

This code needs a CSV file named students.csv and was not run here.

2 Univariate, bivariate and multivariate analysis

Type	Looks at	Examples	Typical charts
Univariate	one variable	mean and spread of marks; distribution of ages	histogram, box plot, bar chart
Bivariate	two variables together	study hours vs marks; correlation of sleep and marks	scatter plot, grouped bars
Multivariate	three or more variables together	correlations of marks, attendance, study hours, sleep	heatmap, pair plot

3 Descriptive statistics

> 💡 **KEY POINT**
> Statistics supports **data collection** and **distribution**; Data Science uses it for **data analysis** and to **understand data patterns**.

Concept	Formula	Explanation	Example
Mean	sum / count	Average value	Average of marks
Median	middle value (sorted)	Value in the middle	Salary with outliers
Mode	most frequent value	Occurs most often	Fashion / surveys
Variance	average of squared deviations	How spread out the data is	Stock volatility
Std deviation	√variance	Average distance from the mean	Test-score spread
Range	max − min	Total spread	9 − 2 = 7

```
import pandas as pd
s = pd.Series([2, 4, 4, 4, 5, 5, 7, 9])
print(s.mean(), s.median(), s.mode()[0], s.max() - s.min())
print(s.var(ddof=0), s.std(ddof=0))
```

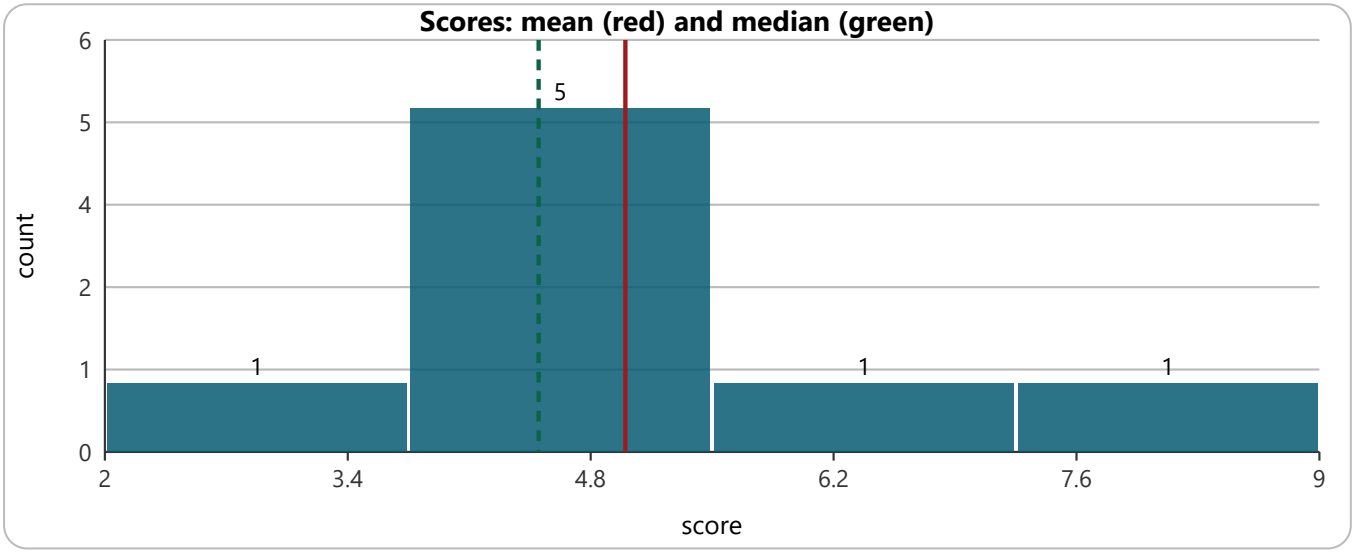
mean 5, median 4.5, mode 4, range 7; population variance 4 and standard deviation 2 (ddof=0 divides by n, like the lecture definition).

اردو مثال

پانچ ملازمین کی تنخواہ 30، 32، 35، 36 ہزار اور مالک کی 400 ہزار ہو تو اوسط (mean) بہت بڑھ جاتا ہے مگر درمیانی قدر (median) تقریباً ٹھیک رہتی ہے۔ اس لیے تنخواہ کے لیے median بہتر ہے۔

4 Spread, quartiles and outliers

1. Mean of 2, 4, 4, 4, 5, 5, 7, 9 = 40 / 8 = **5**
2. Squared deviations: 9, 1, 1, 1, 0, 0, 4, 16 → sum = 32
3. Variance = 32 / 8 = **4**
4. Standard deviation = √4 = **2**
5. Range = 9 − 2 = **7**



KEY POINT

The **mean is sensitive** to extreme values; the **median is robust**. **Quartiles** split sorted data into four parts (Q1, median, Q3); **IQR = Q3 − Q1**. A box plot shows these and the outliers.

NOTE

Population variance divides by **n** (as in the lecture definition). When your data is only a sample, software often divides by **n − 1**.

5 Correlation and causation

KEY POINT

Correlation **r** = Cov(X, Y) / (sx · sy). It runs from **−1** (perfect negative) through **0** (none) to **+1** (perfect positive). It shows **association**, not cause.

|r|	Strength
0.8 – 1.0	very strong
0.6 – 0.8	strong
0.4 – 0.6	moderate
0.2 – 0.4	weak
below 0.2	very weak / none

```
import pandas as pd
hours = pd.Series([1, 2, 3, 4, 5])
marks = pd.Series([50, 55, 65, 70, 80])
print(round(hours.corr(marks), 3))
```

r = 0.993: a very strong positive relationship between study hours and marks.

Notes example: ice-cream sales and drowning both rise in hot weather: correlation exists, causation does not. Hot weather (a **confounder**) drives both.

اردو مثال

گرمی میں آئس کریم بھی زیادہ بکتی ہے اور لوگ تیراکی بھی زیادہ کرتے ہیں، اس لیے ڈوبنے کے واقعات بھی بڑھتے ہیں۔ آئس کریم اور ڈوبنا ساتھ ساتھ بڑھتے ہیں (correlation) مگر ایک دوسرے کی وجہ نہیں (causation نہیں)۔

📝 Mid-exam style questions (Week 6)

What is EDA and why is it done?

Explain mean, median, mode, variance and standard deviation.

Correlation vs causation with an example.

WEEK 7 (TOPIC 7 OF 8)

Data Visualization

Why we visualise data, Matplotlib and Seaborn, which chart to choose, heatmaps and good visualisation practice.

1 Meaning and importance; Matplotlib

💡 KEY POINT

Data visualisation = representing data in graphical forms such as **charts and graphs**. Patterns, trends, outliers and relationships become easy to see and to explain.

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3], [10, 20, 15])
plt.title("Demo")
plt.xlabel("x")
plt.ylabel("y")
plt.show()
```

Draws a line chart; it needs Matplotlib and shows a picture, so it is illustrative here. Key calls from your notes: plt.plot(), plt.bar(), plt.scatter(), plt.show().

2 Choosing a chart

KEY POINT

Choose the chart by the question you are answering.

Chart	Best used for	Example	Poor choice when
Bar chart	Comparing categories	Sales by product	Too many categories
Line chart	Trends over time	Monthly revenue	Unordered categories
Pie chart	Simple part-to-whole	Market share (few slices)	Many or similar slices
Histogram	Distribution of one numerical variable	Age distribution	Categories or two variables
Scatter plot	Relationship between two numbers	Study hours vs marks	Categorical variables
Box plot	Spread and outliers	Salary distribution	Tiny samples
Heatmap	Many values as colours	Correlation matrix	Only a few numbers

اردو مثال

مہینہ وار آمدنی دکھانی ہو تو لائن چارٹ، مختلف دکانوں کا موازنہ ہو تو بار چارٹ، اور پڑھائی کے گھنٹے بمقابلہ نمبر ہو تو اسکیٹر پلاٹ۔

3 Seaborn and heatmaps

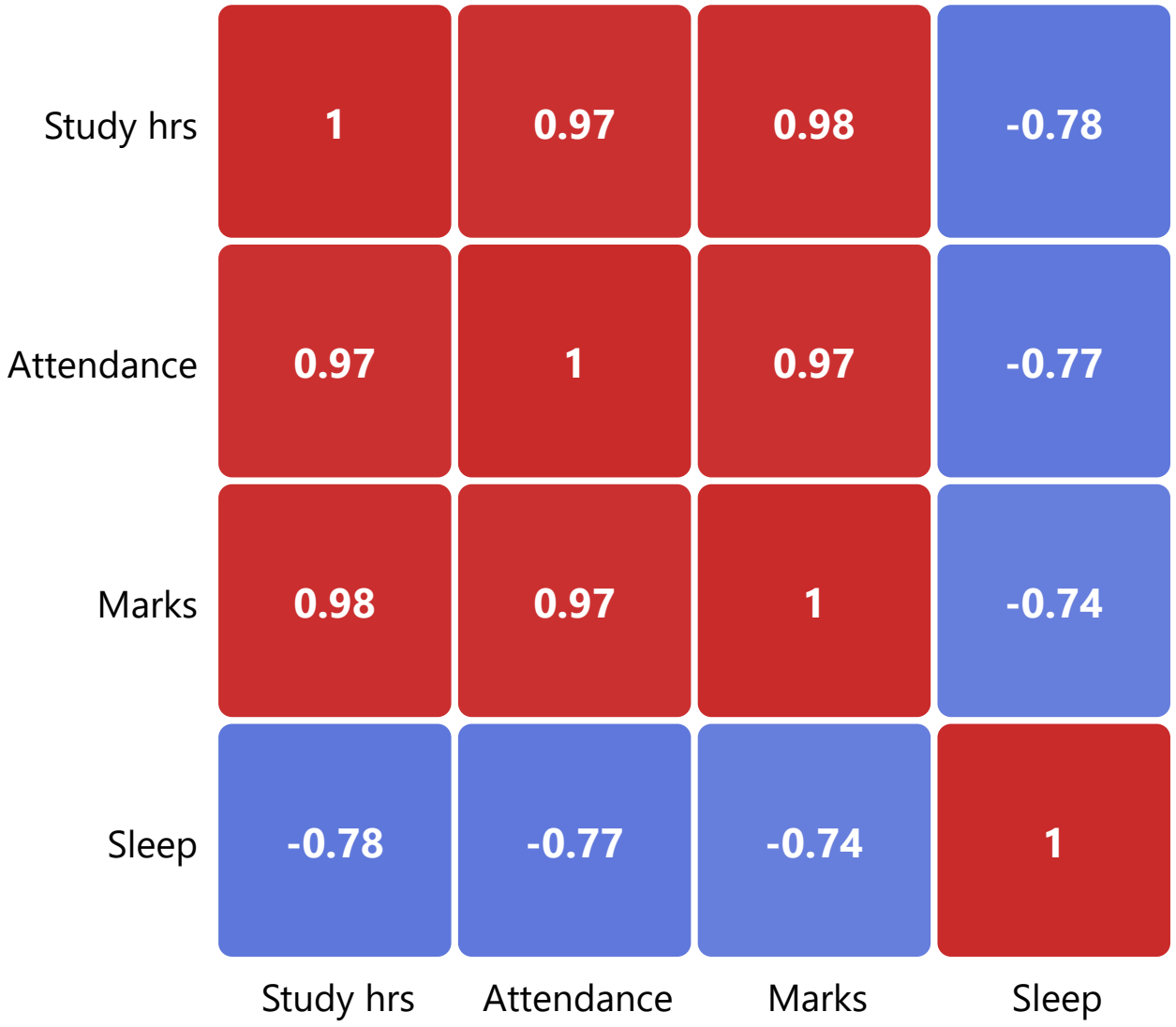
> 💡 **KEY POINT**
> **Seaborn** builds on Matplotlib and makes statistical charts easy: `sns.heatmap()`, `sns.boxplot()`, `sns.pairplot()`.

```
import seaborn as sns
sns.heatmap(df.corr(), annot=True)   # correlation heatmap
sns.boxplot(x=df["salary"])          # spread and outliers
sns.pairplot(df)                     # scatter plots of all pairs
```

Illustrative: needs seaborn and your own DataFrame df.

A **heatmap** shows a table of numbers as colours. For a correlation matrix, each cell is r between two variables; strong positive or negative cells stand out immediately.

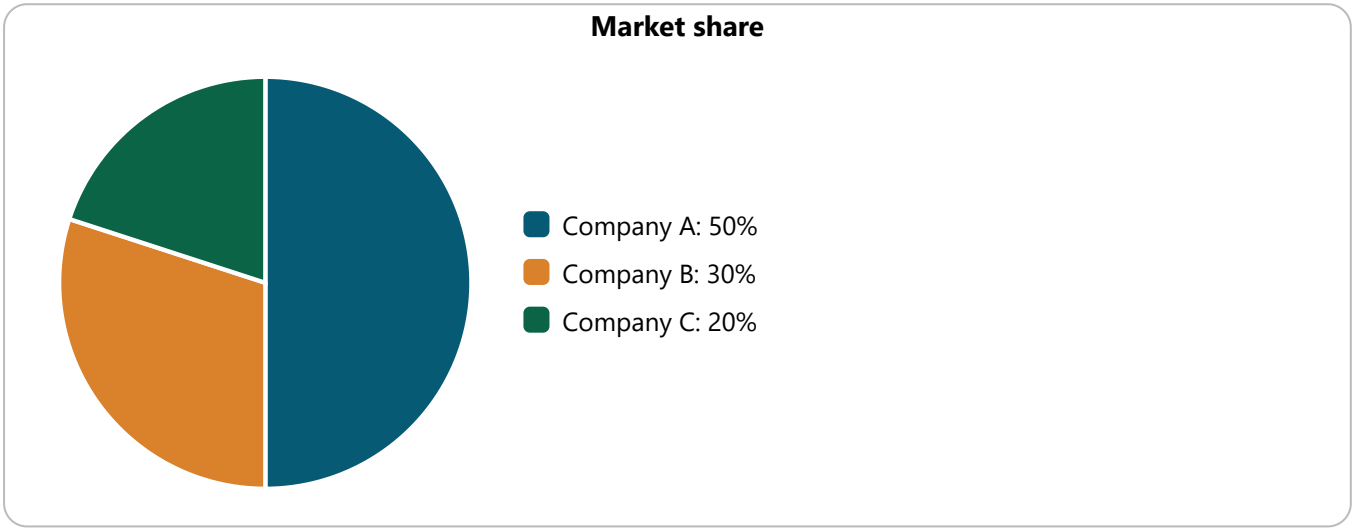
Correlation matrix (example)



red = positive correlation · blue = negative · white = none

4 Visualisation principles

- A clear **title** and labelled **axes** (with units).
- A **legend** when several series are shown.
- Readable **scales**: start bar charts at 0.
- Sensible, limited **colours**; avoid clutter and 3D effects.
- Avoid **misleading** charts: truncated axes, pie charts with many slices.



A pie works here because there are only three parts of one whole.

5 Saving and sharing charts

From your notes: save charts as **images** and write a **PDF report**.

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3], [3, 1, 2])
plt.savefig("chart.png")   # saves the picture to a file
```

Illustrative: run it in your own notebook.

📝 Mid-exam style questions (Week 7)

What is data visualisation and why is it important?

Explain common charts and their uses.

What makes a chart misleading?

WEEK 10 (TOPIC 8 OF 8)

Machine Learning Fundamentals

What ML is, its lifecycle, types of learning, classification vs regression, splitting data, validation, underfitting and overfitting.

1 What is Machine Learning? The ML lifecycle

> 💡 KEY POINT
> **Machine Learning** = algorithms that **learn patterns from data** to make predictions or decisions, instead of following rules written by hand.

1. **Collect** data
2. **Prepare** it (clean, encode, scale)
3. **Split** into training, validation and test sets
4. **Train** the model on the training set
5. **Evaluate / validate** and tune
6. **Deploy** the model
7. **Monitor** and retrain when needed

اردو مثال

اسپیم فلٹر کو ہم ہر اسپیم کا اصول نہیں لکھ کر دیتے؛ ہم ہزاروں ای میلز (اسپیم اور غیر اسپیم) دکھاتے ہیں اور ماڈل خود پیٹرن سیکھ لیتا ہے۔

2 Supervised, unsupervised and reinforcement learning

Type	Data	Goal	Example
Supervised	Labeled: inputs with known answers (targets)	Predict the answer for new data	Spam detection, house-price prediction
Unsupervised	Unlabeled: no answers given	Find structure or groups	Customer grouping
Reinforcement	No fixed dataset: an agent acts and gets rewards	Learn actions that earn the most reward	Robot learning to walk

3 Classification vs regression

> 💡 **KEY POINT**
> In supervised learning the target is either a **category** (classification) or a **number** (regression).

	Classification	Regression
Predicts	A category / label	A continuous number
Examples	Spam or not spam; loan approved or rejected; cat / dog / bird	House price; final marks; temperature

If there is **no target** and you only want groups, that is **clustering** (unsupervised learning).

4 Train / validation / test split

Set	Purpose
Training	The model **learns** patterns from these rows
Validation	Compare models and **tune** settings while building
Test	One final **honest check** on unseen data, used once at the end

```
data = list(range(10))
cut = int(len(data) * 0.8)
train, test = data[:cut], data[cut:]
print(train, test)
```

With scikit-learn: `X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)` (needs the package and your own data).

> ⚠ **NOTE**
> **Data leakage:** if test rows influence training or tuning (for example scaling on all the data, or tuning on the test set), the score looks better than reality. A common split is about 70–80% training with the rest for validation and testing.

5 Generalisation and underfitting

> 💡 **KEY POINT**
> A model learns from **training data**, but its real job is to perform well on **new data it has never seen**. This is **generalisation**. Overfitting and underfitting are two ways it fails.

> 💡 **KEY POINT**
> **Underfitting:** the model is **too simple**. It has not learned the pattern, so it performs **poorly on both** training data and new data (**high bias**).

Causes: too few features, short training, strong regularisation, a model too simple for the pattern.

6 Overfitting

> 💡 **KEY POINT**
> The model is **too complex**. It **memorises the training data including its noise**, so it looks excellent in training but performs poorly on new data (**high variance**).

Causes: too complex a model, too many features, a small dataset, excessive training.

7 Good fit, the student analogy and comparison table

> 💡 **KEY POINT**
> **Good fit:** the model captures the true patterns and performs well on **both** training data and new data.

- **Underfitting** = a student who barely studied.
- **Overfitting** = a student who memorised last year's paper word for word.
- **Good fit** = a student who understands the concepts and can solve any new question.

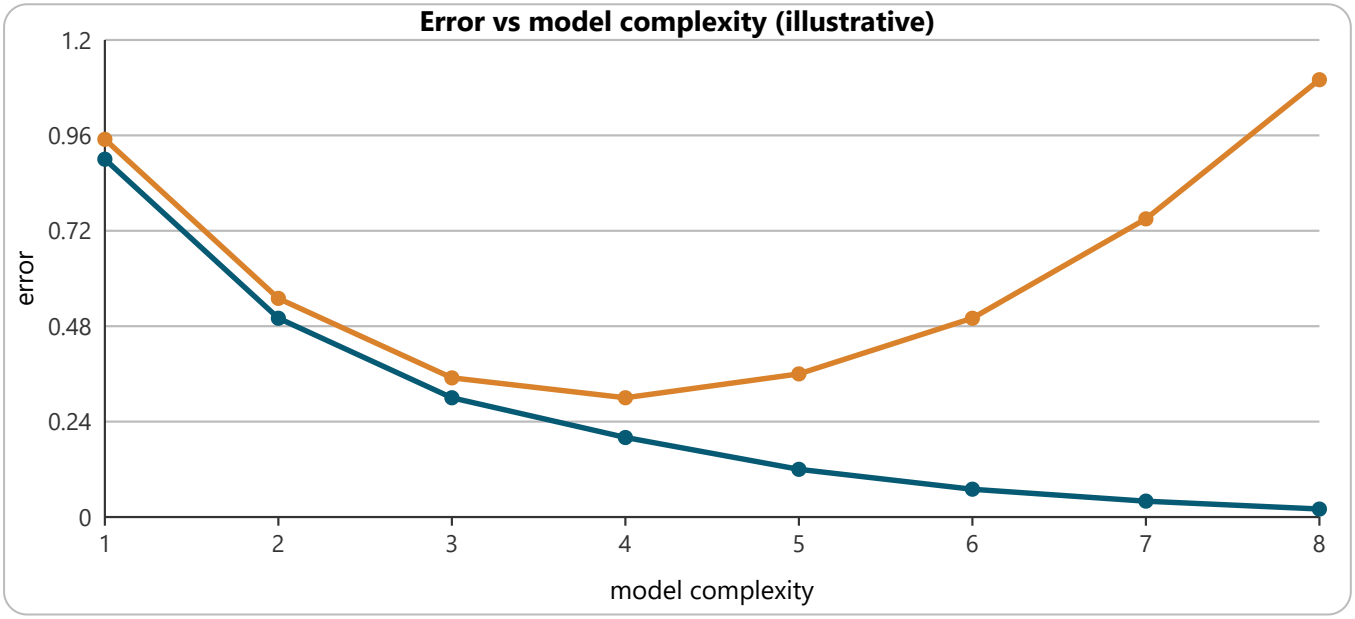
> **اردو مثال**
>
> ایک طالب علم نے پچھلے سال کا پرچہ رٹ لیا (overfitting): پرانے سوال ٹھیک مگر نیا سوال غلط۔
> دوسرے نے بالکل پڑھا ہی نہیں (underfitting): ہر سوال غلط۔ تیسرے نے تصورات سمجھے (good fit): نیا سوال بھی حل کر لیتا ہے۔

Aspect	Underfitting	Overfitting
Meaning	Model too simple to learn the pattern	Model learns pattern plus noise
Also called	High bias	High variance
Training error	High	Very low
Test error	High	High
Gap between train and test	Small	Large
Model complexity	Too low	Too high
Typical cause	Few features, short training, strong regularisation	Small data, too many features
Data needed	More features or a better model may help	More data usually helps
Main fix	Increase complexity, train longer	Simplify, regularise, add data
Learning curve	Both curves high and close together	Training low, validation high, wide gap
Analogy	Student who did not study	Student who memorised answers

8 Quick diagnosis and fixes

Training accuracy	Test accuracy	Diagnosis
58%	55%	**Underfitting**: both are low
98%	70%	**Overfitting**: large gap
90%	88%	**Good fit**: both high and close

Test error is lowest at a **medium** complexity. Beyond that the model starts to overfit:



Problem	Solutions
Underfitting	Use a model with suitable complexity, add useful features, reduce excessive regularisation, train appropriately.
Overfitting	Simpler model, regularisation, cross-validation, feature selection, pruning, early stopping, more representative data.

> ⚠ **NOTE**
> The numbers in the first table are illustrative, as in your notes. In practice judge by both the level and the gap, not a fixed cut-off.

Rule of thumb: **both errors high → add capacity; big gap → reduce complexity or add data**.

📝 Mid-exam style questions (Week 10)

Explain supervised, unsupervised and reinforcement learning with examples.

Why do we split data into training, validation and test sets?

What is overfitting? Give causes and solutions.

What is underfitting? Give causes and solutions.

One-Page Cheat Sheet

Week 1: Introduction

- DS = insight from raw data: statistics + programming + domain knowledge
- AI ⊃ ML; DA = reports and dashboards; DS = whole path from data to insight
- Lifecycle: Problem → Acquire → Explore → Clean → EDA → Features → Model → Evaluate → Deploy → Monitor
- Scientist: models · Analyst: reports · Engineer: pipelines · ML Engineer: deploy

Week 2: Python

- Types: int, float, str, bool
- list [ ] changeable · tuple ( ) fixed · set { } unique · dict {k: v}
- // whole division · % remainder · ** power
- for / while · def … return
- NumPy: np.array(), mean, a * 2 (vectorised); list + joins, array + adds

Week 3: Data acquisition

- Data = raw facts and figures (pictorial, textual, digital)
- Primary: interview, survey, questionnaire · Secondary: repository, database, cloud
- Structured (SQL) · Semi-structured (JSON, XML) · Unstructured (images, text, audio)
- Formats: CSV, Excel, JSON, database, API
- Quality: accuracy, completeness, consistency, validity, uniqueness, timeliness

Week 4: Cleaning

- Missing: delete or impute (mean / median / mode)
- drop_duplicates · standardise text · investigate outliers
- IQR rule: below Q1 − 1.5·IQR or above Q3 + 1.5·IQR
- Nominal → one-hot · Ordinal → label encoding
- Min-max = (x − min)/(max − min) · z = (x − mean)/SD
- Learn scaling values from training data only

Week 5: Pandas

- Series = one column · DataFrame = table
- read_csv, head, info, describe, shape, isnull().sum()
- Filter df[df["marks"] > 70] · sort_values
- groupby("city")["marks"].mean()
- merge on keys (inner / left / right / outer) · join aligns on index

Week 6: EDA

- First look: shape → info → describe → null check
- Univariate (1) · Bivariate (2) · Multivariate (3+)
- Mean = sum/count · Median = middle · Mode = most frequent
- Range = max − min · Variance = average squared deviation · SD = √variance
- r = Cov(X,Y)/(sx·sy), from −1 to +1; correlation ≠ causation

Week 7: Visualization

- Bar categories · Line trend · Pie part-to-whole · Histogram distribution
- Scatter relationship · Box spread/outliers · Heatmap colours
- Matplotlib: plt.plot / bar / scatter / hist / show · Seaborn: heatmap, boxplot, pairplot
- Title, axis labels, legend; start bars at 0

Week 10: ML fundamentals

- ML lifecycle: collect → prepare → split → train → evaluate → deploy → monitor
- Supervised (labels) · Unsupervised (no labels) · Reinforcement (rewards)
- Classification = category · Regression = number · Clustering = groups
- Train: learn · Validation: tune · Test: final check once
- Underfit: both errors high (high bias) · Overfit: train low, test high (high variance) · Good fit: both low and close
- Fix underfit: more complexity / features · Fix overfit: simplify, regularise, more data